1.0 PROJECT OBJECTIVE

The objective of the project is to evaluate SELECT study approaches for options identified in the ASSESS study and identified in ITT SOW and recommend the best option for Concept Design, considering HSE, cost, resources, and schedule and subsequently carry out the Concept Design for the selected approach and prepare DEFINE scope of work for the next stage. In addition, as an optional scope dual concept design study to be carried upon COMPANY confirmation.

The SELECT Study will also address complex project challenges including high UTC, offshore-onshore integration, aging asset replacement, and alignment with ADNOC's broader strategy to maximize oil recovery and gas export while ensuring sustainable and capital-efficient development.

2.0 REFERENCE

2.1 STANDARDS, GUIDELINES & CODES OF PRACTICES

Table. : COMPANY Specifications and Documents

Document Number	Document Title
AGPS-DCoP-GN-01	Design of Upstream and Downstream Processing Systems
AGPS-DCoP-GN-02	Wellhead Tower Design Code of Practice
AGPS-SP-I&C-001	Process Control System Specification
AGPS-SP-I&C-002	Emergency Shutdown (SIS) System Specification
AGPS-SP-I&C-003	Fire & Gas System Specification
ASU-WP2-DS-HQ-WI-IAF-04-Aug2023	ADNOC Smart Upstream Well Instrumentation Guideline

2.2 PROJECT SPECIFICATIONS AND DOCUMENTS

Table 4.2.1: Project Specifications and Documents

Document / Reference Number	Document Title
11536-ML-GRL-J-0001	UMM AL DALKH - PRELIMINARY I&C EQUIPMENT LIST

Document / Reference Number	Document Title
11536-XX-XXX-X-XXX	PLOT PLANS FOR 32MDB SUSTAIN, 60MBD & 100 MBD
11536-A3-GRL-F-3512	BLOCK FLOW DIAGRAM FOR 32 MBD SUSTAIN, 60 MBD & 100 MBD
11536-A3-GRL-G-3503	EQUIPMENT LIST (MULTI-DISCIPLINARY)
C143-000-16-50-LL-0002	ELECTRICAL LOAD LIST (60 MBD & 100 MBD)
11536-ML-GRL-T-0001	TELECOM EQUIPMENT LIST
11536-BD-GRL-G-0003	OVERALL PROJECT DESIGN BASIS
11536-GD-GRL-G-0028	DIGITAL THINKING WORKSHOP REPORT

3.0 PURPOSE OF THIS DOCUMENT

The purpose of this document is to define the adequacy and modification requirements of the existing Instrumentation & Control systems at existing Well head platforms (i.e WHP-01, 02, 03, 04, 07, 11, 13, 14, 15, 16, 19, 21, 22, 24, 32) and associated facilities ~~like~~ such as UAD CPC & Rumaitha Central Control Room -for addition of new signals corresponding to the ~~of new wells as part of UAD LTDP-2~~ 60MBD & 100 MBD development s~~S~~trategies of the UAD LTDP-2 Project~~y~~.

This adequacy assessment has been performed considering the maximum expansion requirements under all strategies of 60 MBD and 100 MBD production cases, including signal additions, WHCP modifications, ICSS expansion, network extension etc.

The adequacy conclusions presented herein are applicable for all strategies of 60 MBD and 100 MBD configurations.

32MBD SUSTAIN adequacy details are addressed under a separate adequacy study report. Refer Doc. No. 11536-SR-GRL-J-0002.

Each platform/facility-wise adequacy details provided in the subsequent sections.~~are mentioned in the below:~~

4.0 ADEQUACY REPORT OF WHP-01

4.1 EXISTING SYSTEM DETAILS:

Based on the site survey observations, the following mentioned Control and Safety Systems are available in the existing WHP-01 which is normally unmanned with remote monitoring and control is from CPC Control room.

Control System / Equipment Details	Make & Model Details	Tag No.	Location
DCS	SEIMENS S7 400	W01-DCS-002	Main Deck, LER
ESD	SEIMENS S7 400	W01-ESD-001	Main Deck, LER
FGS	SEIMENS S7 400	W01-FGP-012	Main Deck, LER
WHCP	Boustead Controls	W01-Z-001	Main Deck, Outdoor

4.2 OBJECTIVE:

As part of UAD LTDP-2 60MBD and 100MBD development strategies, 4 No's of new ESP OP wells will be added to the existing WHP-01. The addition of these four wells is common across all evaluated 60 MBD and 100 MBD strategies.

The objective is to perform the I&C adequacy of existing DCS/ESD/FGS and WHCP for I&C
modifications required which includes the feasibility to utilise the existing spares and /or necessity of extension of existing DCS/ESD/FGS/WHCP systems to meet the 60MBD & 100
MBD scope new signals

- ~~for I&C modifications required which includes the feasibility to utilise the existing spares and /or necessity of extension of existing DCS/ESD/FGS/WHCP systems to meet the 60MBD & 100 MBD scope new signals.~~

4.3 EXISTING SYSTEM MODIFICATIONS:

Based on Site survey observations and study inputs, adequacy details of each system DCS, ESD, FGS and WHCP are elaborated below with estimated I/O Quantities, MTO, Conclusion and shutdown requirements. Refer site survey report-1 Doc. no. 11536-MR-GRL-G-0002 for site survey details and Process Block Flow Diagram Dwg. No. 11536-A3-GRL-F-3512 for new wells details.

4.3.1 DCS SYSTEM:

Based on the site survey observations, WHP-01 has one (1) no's of the existing DCS cabinet (W01-DCS-002) which is inadequate due to space constraints and limited no. of spare entries availability based on ongoing interface project P11708 - UAD LTDP-1 site survey report. Therefore, New DCS cabinet is envisaged and it will get extended from the existing WHP-01 DCS System network for the new I/O requirements mentioned in the below table.

Note: - One (1) no's of new DCS panel will be installed as part of ongoing interface project. P11708 - UAD LTDP-1 scope for 3 Nos new well signals. The same new cabinet(W01-DCS-03) spare utilisation shall be verified during the next stage of the project prior to adding new DCS cabinet considered for LTDP-2 Project.

New Cabinet will be installed at outdoor suitable for Zone 2, Gr. IIA, T3 min Hazardous area classification. The I/O modules and other components in panel shall comply with Zone 2, Gr. IIA, T3 min based on latest Wellhead tower Design code of practice Doc No. AGPS-DCoP-GN-02.

The existing WHP-01 DCS system communicates with the CPC Control room via the existing redundant FO links through composite cables for remote operation and monitoring from the CPC control room.

For the addition of new 4 Nos of ESP Wells in the WHP-01, the following minimum DCS I/Os are estimated based on Typical existing P&IDs.

Table : WHP-01 - DCS I/O summary (4 Nos of new wells)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	24	5	29	4	
2	AO	4	1	5	1	
3	DI	72	15	87	3	
4	DO	32	6	38	2	

4.3.2 ESD SYSTEM:

Based on the site survey observations, WHP-01 has one (1) no's of the existing ESD cabinet (W01-ESD-001) which is inadequate due to space constraints and limited no. of spare entries availability based on ongoing interface project P11708 - UAD LTDP-1 site survey report. Therefore, New ESD cabinet is envisaged and it will get extended from the existing WHP-01 ESD System network for the new I/O requirements mentioned in the below table.

New Cabinet will be installed at outdoor suitable for Zone 2, Gr. IIA, T3 min Hazardous area classification. The I/O modules and other components in panel and I/O Modules shall comply with Zone 2, Gr. IIA, T3 min based on latest Wellhead tower Design code of practice Doc No. AGPS-DCoP-GN-02.

The existing WHP-01 ESD system is serially interfaced with the existing DCS System for remote operation and monitoring from the CPC Control room and shutdown related logic signals / critical signals are connected hardwired to the ESD system.

For the addition of new 4 Nos of ESP Wells in the WHP-01, the following minimum ESD I/Os are estimated based on Typical existing P&IDs.

Table : WHP-01 - ESD I/O summary (4 Nos of new wells)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	24	5	29	4	
2	AO	-	-	-		
3	DI	32	6	38	2	
4	DO	48	10	58	2	

4.3.3 FGS SYSTEM:

Based on the site survey observations, WHP-01 has one (1) no's of the existing FGS cabinet (W01-FGS-001) which is inadequate due to space constraints and limited no. of spare entries availability based on ongoing interface project P11708 - UAD LTDP-1 site survey report. Therefore, New FGS cabinet is envisaged and it will get extended from the existing WHP-01 FGS System network for the new I/O requirements mentioned in the below table.

New Cabinet will be installed at outdoor suitable for Zone 2, Gr. IIA, T3 min Hazardous area classification. The I/O modules and other components in panel shall comply with Zone 2, Gr. IIA, T3 min based on latest Wellhead tower Design code of practice Doc No. AGPS-DCoP-GN-02.

The existing WHP-01 FGS system is serially interfaced with the existing DCS System for remote operation and monitoring from the CPC Control room and shutdown related logic signals / critical signals are connected hardwired to the DCS & ESD system.

For the new wells additions and platform extensions in the WHP-01, the following minimum F&G I/Os are estimated below. This shall be further studied during the next stage of the project.

Table : WHP-01 - FGS I/O summary (5 Nos of new wells)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	6	2	8	1	
2	AO	-	-	-		
3	DI	5	1	6	1	
4	DO	10	2	12	1	

4.3.4 WHCP:

The OEM of existing Wellhead Control Panel (WHCP) at WHP-01 is M/s Boustead Controls.

WHCP(W01-Z-001) is located in the Main Deck. The existing WHCP is comprised in a single panel with all Hydraulic sections, Well Control Modules and Master Modules with external accumulators arrangement.

During the site survey, it is observed that spare WCM slots are available for future ESP wells.

Existing WHCP is sized to accommodate all the 12 Wells which includes Accumulators, reservoirs, Pump/Motor and JB etc based on LTDP-1 Instrument adequacy report for WHCP. Refer Company document no. 11708-SR-WHP-J-0001 Rev A2 Sec 13.0.

As part of UAD LTDP-2 60MBD and 100MBD scope, 4 No's of new ESP wells are adding to the existing WHP-01 based on Process Block Flow Diagram. Accordingly, 4 No's of spare WCMs are required in the existing WHCP to hook-up with the new ESP wells. Refer below Table summary for WCM spare status including LTDP-1 wells and refer site survey report doc. no. 11536-MR-GRL-G-0002 Sec 7.4.2 for more details.

Below is the table summary of WCMs.

Description	Well Modules (in use) (including LTDP-1, Note-1)	Well modules spare available (Installed spare / Blank slots)	Required for UAD LTDP-2	Spare available after LTDP-2 (Option-2 Scenario)
Well Modules	07	05 (02/03)	04	01

Identified two (2) design options to accommodate 4 Nos of new ESP Wells. The options are mentioned below:

Option-1: New extended WHCP sized for new 4 ESP wells (Refer below Note-2)

Pros:

- Extremely reducing the modification works in the existing WHCP.
- Major reduction in the Wellhead control system downtime.
- Readily installed & future spare slots.

Cons:

- New WHCP cabinet cost
- New foot print and weight

Option-2: Utilisation of existing WHCP for new 4 ESP wells (Refer below image)

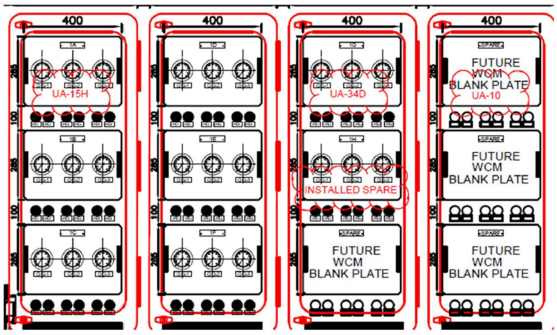
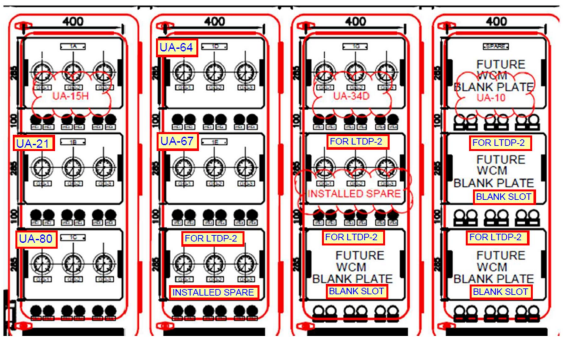
Pros:

- Less CAPEX, avoid new wellhead control panel extension.
- Existing installed WCMs can be utilized for new wells.
- Avoid new foot print and weights.
- Avoid Hydraulic header tubing tie-in modification works.

Cons:

- Zero installed WCMs spares. However, Loose supply WCMs shall be considered to maintain the spares availability.
- More shutdown time is required for existing panel modifications and testing of WCMs.
- More internal Installation / Modifications of Tubing & wiring

Refer below snapshot extracted from WHCP adequacy report of UAD LTDP-1 interface project (P11708). Refer to WHCP adequacy report Dwg No. 11708-SR-WHP-J-0001.

Existing WHCP (Including LTDP-1)	Existing WHCP for LTDP-2 (For above Option-2 scenario)
	

2 Nos of Installed spare WCM and 2 Nos of blank slots will be utilised for LTDP-2 Project.

Based on interface Project No. P11708 (UAD LTDP-1) WHCP adequacy outcomes that existing WHCP's reservoir, HP & MP Accumulator, Hydraulic pumps, is designed for 12 no's of Wells. Therefore, no modifications are envisaged for the reservoir, accumulators & pumps for addition of 4 no's of wells. However, OEM adequacy shall be verified during the FEED stage with the FEED stage actual process and site conditions.

Notes: -

1. 3 Nos of new ESP wells (UA-15H, UA34D and UA-10) are added as part of LTDP-1 project. The existing ESP Wells are UA-21, UA-80, UA-64 & UA-67.

2. New extended WHCP shall be sized with 6 Nos of WCMs for LTDP-2 scope wells. The existing WHCP installed spare WCM (2 Nos) and blank WCM lots (3 Nos) will be retained as spares.

4.4 CONCLUSION:

a) ICSS:

Based on SELECT stage adequacy outcomes, existing DCS, ESD & FGS cabinets are not adequate to cater new I/Os. Therefore, the following items and modifications are required in the existing WHP-01 control system and further integration with existing CPC.

S.No	New Item Description / Modifications	Quantity	Remarks
1	New DCS Cabinet - MOC: SS316L, IP66 min., suitable for Zone 2, IIB, T3 Min, Outdoor Installation. - New I/O Rack with Interface Modules, I/O Modules for extending to an existing DCS). - New Terminal Blocks, etc.	1 Nos	
2	New ESD Cabinet - MOC: SS316L, IP66 min., suitable for Zone 2, IIB, T3 Min. - New I/O Rack with Interface Modules, I/O Modules for extending to an existing DCS). - New Terminal Blocks, etc.	1 Nos	
3.	New FGS Cabinet - MOC: SS316L, IP66 min., suitable for Zone 2, IIB, T3 Min. - New I/O Rack with Interface Modules, I/O Modules for extending to an existing FGS). - New Terminal Blocks, etc.	1 Nos	
4.	Existing DCS/ESD/F&G Cabinets modifications: - Hardware Interface with new system cabinets - Software updations of new tag configurations, graphics updations etc	1 Lot	

b) WHCP:

Based on SELECT stage adequacy outcomes, Option-1 is recommended to avoid major modifications in the existing WHCP and to carried with minimal downtime and minimal tie-in modifications of MP & HP hydraulic headers conditions. During next stage, detailed optimized option shall be verified with the OEM and Company operations for maintenance spares requirements. The following items and modifications are required as minimum as to hook-up with New 4 ESP wells based on OPTION-1:

S.No	New Item Description / Modifications	Quantity	Remarks
1	New Well Head Control Panel (Extension Cabinet for New Modules only) - MOC: SS316L, min. IP66, suitable for Zone 1, IIB, T3 min. - New 4 no. of Modules, JB etc for new wells etc.	1 Nos	
2	HP & MP Hydraulic Tubing header tie-in modifications	1 Lot	

4.5 SHUTDOWN REQUIREMENTS:

The shutdown is envisaged to carry out the above modifications. However, this shall be reviewed during the FEED stage information.

5.0 ADEQUACY REPORT OF WHP-02

5.1 EXISTING DETAILS:

Based on the site survey observations, the following mentioned Control and Safety systems are available in the existing WHP-02 which is normally unmanned with remote monitoring and control is from CPC Control room.

Control System / Equipment Details	Make & Model Details	Tag No.	Location
DCS	SEIMENS S7 400	W02-DCS-002	Main Deck, Indoor (LER)

ESD	SEIMENS S7 400	W02-ESD-001	Main Deck, Indoor (LER)
FGS	SEIMENS S7 400	W02-FGS-012	Main Deck, Indoor (LER)

5.2 OBJECTIVE:

As part of UAD LTDP-2 60 and 100MBD scope, 2 No's of new ESP wells will be added to the existing WHP-02.

The objective is to perform the I&C adequacy of existing DCS/ESD/FGS and WHCP for I&C modifications required which includes the feasibility to utilise the existing spares and /or necessity of extension of existing DCS/ESD/FGS/WHCP systems to meet the 60MBD & 100 MBD scope new signals.

5.3 EXISTING SYSTEM MODIFICATIONS:

Based on Site survey observations and study inputs, adequacy details of each DCS, ESD and WHCP are elaborated in below with I/O Quantities, MTO, Conclusion and shutdown requirements. Refer site survey report-2 Doc. no. 11536-MR-GRL-G-0003 and Process Block Flow Diagram Dwg. No. 11536-A3-GRL-F-3512 for well details.

5.3.1 DCS SYSTEM:

Based on the site survey observations, WHP-02 has one (1) no's of the existing DCS cabinet (W02-DCS-002) installed inside the LER. Another one (1) no. of new DCS Cabinet will be installed outside the LER as part of ongoing interface project P11708 - UAD LTDP-1.

Based on Interface project P11708 Inst Adequacy study report drawing No. 11708-SR-GRL-J-0002_A2, The new DCS Panel installed under LTDP-1 will be having spare space to install 1 Nos of new I/O Rack. Therefore, New I/Os addition shall be in the New DCS cabinet being added under UAD LTDP-1 project by others.~~the modification in this existing UAD LTDP-1 DCS cabinet is envisaged to accommodate the below mentioned new I/O's.~~

High level desktop adequacy for DCS cabinet has been checked based on available drawings from interface project. However detailed adequacy shall be performed during FEED stage by OEM (SIEMENS). Accordingly, required hardware shall be considered for the new I/Os.

Table : WHP-02 - DCS I/O summary (New 2 No. OP Well)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	12	3	15	2	
2	AO	2	1	3	1	
3	DI	36	8	44	2	
4	DO	16	4	20	1	

5.3.2 ESD SYSTEM:

Based on the site survey observations, WHP-02 has one (1) no's of the existing ESD cabinet (W02-ESD-001) which is inadequate due to space constraints and limited no. of spare entries availability based on ongoing interface project P11708 - UAD LTDP-1 site survey report. Therefore, New ESD cabinet is envisaged and it will get extended from the existing WHP-02 ESD System network for the new I/O requirements mentioned in the below table.

New Cabinet will be installed at outdoor suitable for Zone 2, Gr. IIA, T3 min Hazardous area classification. The I/O modules and other components in panel and I/O Modules shall comply with Zone 2, Gr. IIA, T3 min based on latest Wellhead tower Design code of practice Doc No. AGPS-DCoP-GN-02.

The existing WHP-02 ESD system is serially interfaced with the existing DCS System for remote operation and monitoring from the CPC Control room and shutdown related logic signals / critical signals are connected hardwired to the ESD system.

For the addition of new 2 Nos of ESP Wells in the WHP-02, the following minimum ESD I/Os are estimated based on Typical existing P&IDs.

Table : WHP-02 - ESD I/O summary (New 2 No. OP Well)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	12	3	15	2	

2	AO	2	1	3	1	
3	DI	36	8	44	2	
4	DO	20	4	24	1	

5.3.3 FGS SYSTEM:

Based on the site survey observations, WHP-02 has one (1) no's of the existing FGS cabinet (W02-FGP-012) which is inadequate due to space constraints and limited no. of spare entries availability based on ongoing interface project P11708 - UAD LTDP-1 site survey report. Therefore, New FGS cabinet is envisaged and it will get extended from the existing WHP-02 FGS System network for the new I/O requirements mentioned in the below table.

New Cabinet will be installed at outdoor suitable for Zone 2, Gr. IIA, T3 min Hazardous area classification. The I/O modules and other components in panel shall comply with Zone 2, Gr. IIA, T3 min based on latest Wellhead tower Design code of practice Doc No. AGPS-DCoP-GN-02.

The existing WHP-02 FGS system is serially interfaced with the existing DCS System for remote operation and monitoring from the CPC Control room and shutdown related logic signals / critical signals are connected hardwired to the DCS & ESD system.

For the new wells additions and platform extensions in the WHP-02, the following minimum F&G I/Os are estimated below. This shall be further studied during the next stage of the project.

Table: WHP-02 - FGS I/O summary (New 2 No. OP Well)

S.No	I/O TYPE	Actual I/O Qty	20% Spare I/O Qty	Total I/O Qty	Total New I/O Modules	Remarks
1	AI	06	2	08	1	
2	AO	0	0	0	0	
3	DI	05	1	06	1	
4	DO	10	2	12	1	

5.3.4 WHCP:

The OEM of existing Wellhead Control Panel (WHCP) at WHP-02 is M/s Boustead Controls.

WHCP(W02-Z-001) is located in the Main Deck. The existing WHCP is comprised in a single panel with all Hydraulic sections, Well Control Modules and Master Modules with external accumulators arrangement.

During the site survey, it is observed that spare WCM slots are available for future ESP wells.

The existing WHCP is sized to accommodate 12 Wells which includes accumulators, reservoirs, Pump/Motor and JB etc. based on LTDP-1 Instrument adequacy report for WHCP. Refer Company document no. 11708-SR-WHP-J-0001 Rev A2 Sec 13.2

As part of UAD LTDP-2 60MBD & 100MBD, 2 No's of new ESP wells are adding to the existing WHP-02 based on Process Block Flow Diagram. Accordingly, 2 No's of spare WCMs are required in the existing WHCP to hook-up with the new ESP wells. Refer below Table summary for WCM spare status including LTDP-1 wells and refer site survey report doc. no. 11536-MR-GRL-G-0003 Sec 7.2.4 for more details.

Below is the table summary for the WCMs.

Description	Well Modules (in use) (including LTDP-1, Note-1)	Well Modules spare slots (Installed spare / Blank slots)	Required for UAD LTDP-2	Spare available after LTDP-2
Well Modules	07	05 (04/01)	02	03

Based on interface Project No. P11708 (UAD LTDP-1) WHCP adequacy outcomes that existing WHCP's reservoir, HP & MP Accumulator, Hydraulic pumps, is designed for 12 no's of Wells. Therefore, no modifications are envisaged for the reservoir, accumulators

& pumps for addition of 2 no's of wells. However, OEM adequacy shall be verified during the FEED stage with the FEED stage actual process and site conditions.

Notes: -

1. 1 Nos of new ESP wells (UA-18D) are added as part of LTDP-1 project. The existing ESP Wells are UA-19, UA-18H, UA-59, UA-62, UA-41. UA-76 and existing water disposal wells are WD-1 & WD-2.

5.4 CONCLUSION:

a) ICSS:

Based on SELECT stage adequacy outcomes, existing DCS cabinet is not adequate, however the new DCS panel being added in UAD LTDP-1 project is considered to be adequate for UAD LTDP-2 project I/Os addtions ~~existing DCS cabinet is adequate~~ and existing ESD & FGS cabinets are inadequate to cater new I/Os. Therefore, the following items and modifications are required in the existing WHP-02 control system and further integration with existing CPC.

S.No	New Item Description / Modifications	Quantity	Remarks
1	New ESD Cabinet - MOC: SS316L, IP66 min., suitable for Zone 2, IIB, T3 Min. - New I/O Rack with Interface Modules, I/O Modules for extending to an existing DCS). - New Terminal Blocks, etc.	1 Nos	
2	New FGS Cabinet - MOC: SS316L, IP66 min., suitable for Zone 2, IIB, T3 Min. - New I/O Rack with Interface Modules, I/O Modules for extending to an existing FGS). - New Terminal Blocks, etc.	1 Nos	

3	Existing DCS/ESD/F&G Cabinets modifications: - Hardware modifications for new I/Os addition in the ESD & FGS Cabinets. - Hardware Interface with new system cabinets - Software updations of new tag configurations, graphics updations etc	1 Lot	

b) WHCP:

Based on SELECT stage adequacy outcomes, following items and modifications are required as minimum as to hook-up with New 2 ESP wells.

S.No	New Item Description / Modifications	Quantity	Remarks
1	Existing Well Head Control Panel Modifications - Configuration / Hook-up the WCM with new wells - Bulk Head tubing installations / modifications etc	1 Lot	
2	HP & MP Hydraulic Tubing header tie-in modifications	1 Lot	

5.5 SHUTDOWN REQUIREMENTS:

The shutdown is envisaged to carry out the above modifications. However, this shall be reviewed during the FEED stage information.